

EEE 416 (January 2025)
Microprocessor and Embedded Systems Laboratory

Final Project Report

Section: A1 Group: 07

**Smart Performance Monitoring System
in Agricultural Farm**

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1 Abstract

The agricultural sector in Bangladesh, while a cornerstone of the national economy, largely operates on traditional practices, limiting its potential to fully leverage modern technological advancements. This reliance on conventional methods often results in subjective decision-making, leading to significant post-harvest crop wastage and a lack of data-driven insights into farm performance. To bridge this technological gap and enhance agricultural efficiency, this project introduces a "Smart Performance Monitoring System in Agricultural Farms." Built on an ESP32 microcontroller, the system integrates a low-cost, non-destructive sensor suite—including load cells, gas sensors (for ethanol, methane, hydrogen, and ammonia), and environmental sensors—to collect real-time data on crop yield quality and storage conditions. By analysing this data to generate a comprehensive quality index, the system provides immediate feedback to farmers via an OLED display and logs detailed information to a cloud-based dashboard. This solution has the potential to empower farmers of our country with the tools for informed decision-making, aiming to significantly reduce crop spoilage, improve yield quality, and pave the way for more sustainable and productive farming practices.

2 Introduction

The agricultural sector is a vital component of Bangladesh's economy, contributing approximately 13.6% to the national GDP and employing over 40% of the workforce (BBS, 2023). Despite its significance, the sector remains heavily reliant on traditional farming practices, characterized by manual labor and experiential knowledge passed down through generations. While these methods hold cultural and historical value, they often fail to leverage modern technological advancements, leading to systemic inefficiencies. A critical issue originating from this technological lag is the reliance on subjective decision-making in crop management and post-harvest handling, which contributes to substantial post-harvest losses, estimated at 20-30% of produce in developing economies like Bangladesh (FAO, 2020). These losses, driven by inadequate monitoring of crop storage conditions, pose a significant threat to food security and economic stability for rural farmers.

To address this challenge, our project introduces an innovative, cost-effective technological solution designed specifically for rural Bangladeshi farming communities. The system aims to shift agricultural practices from intuition-based approaches to data-driven precision agriculture, empowering farmers to make informed decisions. By focusing on early detection of spoilage through indicators such as metabolic gas emissions, the system provides continuous, non-destructive monitoring of crop health during storage. This approach not only mitigates losses but also enhances resource efficiency, aligning with sustainable agricultural practices.

The system is based on the ESP32 microcontroller that integrates a comprehensive sensor suite to capture critical environmental and physical data. This includes a load cell-based weighting platform for accurate measurement of crop quantities, gas sensors (MQ-3, MQ-4, MQ-8, and MQ-135) to detect volatile compounds indicative of spoilage such as alcohol, methane, ammonia, hydrogen, and a DHT22 sensor to monitor temperature and humidity—key factors influencing post-harvest deterioration (Kader, 2002). An RFID module further enhances functionality by enabling precise tracking and identification of produce batches, as well as farmers, improving traceability across the supply chain. Real-time data is displayed on an OLED screen for immediate on-field decision-making, while wireless connectivity to a cloud-based Google Sheet dashboard ensures long-term data storage, enabling trend analysis and historical review to optimize future farming strategies.



Figure: United Nation's Sustainable Development Goals

This project also incorporates a user-centric design, prioritizing affordability and ease of use to ensure accessibility for resource-constrained farmers. By leveraging open-source technologies and low-cost components, the system addresses the economic barriers often faced by rural communities in adopting advanced agricultural tools. Furthermore, it contributes to global sustainability goals, particularly the United Nations' Sustainable Development Goal 2 (Zero Hunger), by promoting efficient resource use and reducing food waste. Ultimately, the "Smart Performance Monitoring System" seeks to bridge the technological divide in Bangladesh's agricultural sector, fostering a more resilient, productive, and economically sustainable future for rural farmers. The development of a smart performance monitoring system in agricultural farms to mitigate post-harvest losses in rural Bangladesh represents a complex engineering challenge due to its interdisciplinary nature, resource constraints, and socio-cultural considerations. This problem requires integrating embedded systems, sensor technology (gas, temperature, humidity, and load sensors), wireless communication, and data analytics to provide real-time, non-destructive crop quality assessment. The system must be affordable, energy-efficient, and durable in harsh environmental conditions (e.g., 20–40°C, 70–90% humidity), while remaining user-friendly for farmers with limited technical literacy. Scalability across diverse crops and adaptability to evolving needs further complicate the design, as does the need for robust data processing and

reliable cloud connectivity in areas with inconsistent infrastructure. Alternative solutions include solar-powered cold storage, which is effective but very costly and power-dependent for rural farmers; chemical preservatives, limited by health concerns and crop specificity; centralized IoT systems, which offer advanced analytics but require expensive infrastructure and reliable internet; and mobile app-based advisories, constrained by manual input and digital literacy requirements. The proposed ESP32-based system balances cost, automation, and accessibility, making it suitable for rural contexts.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The project aims to develop a low-cost, real-time monitoring solution to reduce post-harvest losses for rural Bangladeshi farmers by providing data-driven insights into crop quality and storage conditions. Since it would be used by rural farmers, there are certain constraints in designing and implementing it. They include:

- Low cost ensuring affordability by rural farmers
- Low powered to be operated in rural conditions
- Low maintenance and requirement of less technological knowledge
- Durability in harsh conditions

Meeting all these constraints while delivering a good quality monitoring system is a challenging task. Our prototype for this includes designing an ESP32 microcontroller-based system with sensors (load cell, MQ-3, MQ-4, MQ-8, MQ-135, DHT22, RFID) to monitor crop quantity, gas emissions, temperature, humidity, and farmer tracking, featuring real-time OLED display and wireless data transmission to a Google Sheet dashboard for analysis, all while targeting affordability and durability in humid, tropical conditions for crops like rice, vegetables, and fruits, with user-friendly interfaces and basic training materials. Out-of-scope elements include physical storage infrastructure, advanced technologies like hyperspectral imaging or machine learning beyond basic data logging, supply chain management, large-scale or urban deployment, and ongoing maintenance. Deliverables encompass a functional prototype, design documentation, with tasks including needs assessment, hardware-software integration, cloud setup, and farmer training, constrained by component availability, limited rural internet, and technical literacy, all within a seven week timeline.

3.1.2 Literature Review

Crops, once harvested and transported to storage facilities, carry residual moisture, field debris, and microorganisms such as fungi (e.g., *Aspergillus* and *Penicillium*), bacteria, and yeasts (Kader, 2002). These elements, combined with environmental factors like humidity,

temperature, and ventilation, significantly influence the rate of decomposition. High moisture content in crops creates an optimal environment for microbial proliferation, as it facilitates the growth of bacteria and fungi that metabolize starches, sugars, and proteins (Jay, 2012). This microbial activity produces metabolic byproducts such as carbon dioxide, methane, ammonia, hydrogen, and, in some cases, mycotoxins, which further accelerate spoilage and pose health risks (Pitt & Hocking, 2009). For instance, molds like *Aspergillus flavus* can produce aflatoxins, compromising both crop safety and marketability (FAO, 2020).

The storage environment plays a critical role in either exacerbating or mitigating these processes. High humidity (70–90%) and warm temperatures (20–40°C), common in tropical climates like Bangladesh, create favorable conditions for microbial growth, while poor ventilation exacerbates moisture retention (Kader, 2002). Conversely, proper storage practices, such as maintaining low temperatures and controlled humidity, can slow respiration rates and microbial activity, preserving crop quality for months or even years (Thompson, 2010). These findings underscore the need for effective monitoring and management strategies to minimize post-harvest losses.

Traditional Post Harvest Handling and Storage Practices:

Effective post-harvest management begins with determining the maturity indicators such as oil content, dry matter, moisture content, sugar levels (measured via refractometry), starch content, and pH changes which are widely used to assess harvest readiness (Reid, 2002). For example, fruits harvested at peak maturity exhibit lower respiration rates, extending shelf life (Kader, 2002). Post-harvest storage practices, such as refrigeration and controlled atmosphere storage (CAS), are standard in developed countries to slow respiration and microbial deterioration. CAS modulates oxygen and carbon dioxide levels to create an environment less conducive to spoilage, with studies showing a 50–70% reduction in spoilage for fruits like apples and pears (Thompson, 2010). However, these technologies are often cost-prohibitive (e.g., \$1,000–\$5,000 for small-scale refrigeration units) and require consistent power, limiting their adoption in resource-constrained regions like rural Bangladesh (FAO, 2020).

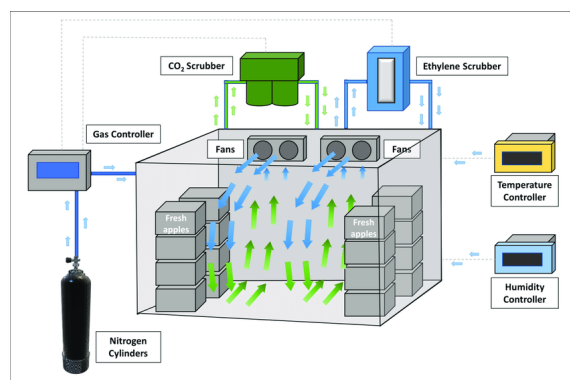


Figure: Controlled Atmospheric Storage (CAS)

Source: Guan, Jiewen, et al. "A review: Gaseous interventions for Listeria monocytogenes control in fresh apple cold storage." Frontiers in Microbiology 12 (2021): 782934.

Alternative low-cost methods, such as hermetic storage, have gained traction in developing countries. Hermetic storage bags or silos create airtight environments that limit oxygen availability, reducing microbial and pest activity. Research indicates that hermetic storage can reduce grain losses by up to 20% in sub-Saharan Africa and South Asia (Baributsa et al., 2017). However, these systems lack real-time monitoring capabilities and are less effective for highly perishable crops like fruits and vegetables, which require active environmental control (FAO, 2020).

Emerging Technologies for Crop Quality Assessment

Recent advancements in non-destructive technologies have revolutionized post-harvest quality assessment, offering precise and scalable solutions. Imaging techniques, such as hyperspectral and multispectral imaging, enable the detection of early spoilage indicators, including changes in color, texture, and chemical composition, without damaging crops (Lu et al., 2016). Spectroscopy, particularly near-infrared (NIR) spectroscopy, has been widely adopted for assessing internal quality parameters like sugar content, moisture, and starch levels in fruits and vegetables. Studies demonstrate that NIR spectroscopy can predict shelf life with 85–90% accuracy for crops like tomatoes and mangoes (Nicolai et al., 2007).

Machine learning (ML) models have further enhanced the predictive power of these technologies. By analyzing data from sensors, imaging, or spectroscopy, ML algorithms can forecast spoilage trends and optimize storage conditions. For instance, convolutional neural networks (CNNs) applied to hyperspectral images have achieved up to 95% accuracy in detecting early fungal contamination in grains (Zhang et al., 2020). However, these technologies require significant computational resources and expertise, making them less accessible for smallholder farmers in developing countries.



Figure: Portable Fruit Quality Meter

Source: Spectroscopy in Agriculture by Felix Instruments

Gas sensor-based systems offer a promising alternative for real-time spoilage detection. Volatile organic compounds (VOCs) emitted during microbial decomposition, such as carbon dioxide, methane, and ammonia, serve as reliable spoilage indicators (Loutfi et al., 2015). Sensors like MQ-series gas sensors (e.g., MQ-3, MQ-4, MQ-8, MQ-135) are cost-effective and can be integrated with microcontrollers like the ESP32 for continuous monitoring. Research shows that VOC-based monitoring systems can detect spoilage in fruits and vegetables within 24–48 hours of onset, enabling timely interventions (Sanaeifar et al., 2017). These systems are particularly suitable for resource-constrained settings due to their low cost (\$20–\$50 per unit) and minimal power requirements. However, challenges remain in ensuring system durability in harsh environmental conditions (e.g., high humidity, dust) and providing adequate training for farmers with limited technical exposure. Combining sensor-based systems with complementary low-cost solutions, such as hermetic storage or mobile app-based advisories, could further enhance their impact (GSMA, 2022). For instance, mobile apps leveraging Bangladesh’s 70% smartphone penetration could provide storage best practices and weather-based alerts, though they rely on manual input and network connectivity (GSMA, 2022).

Microbial Interactions and Gas Production Pathways

The decomposition of harvested crops, particularly fruits and vegetables, is a significant contributor to post-harvest losses, with microbial activity playing a central role in the process. Understanding the interactions between microbes and crop nutrients is critical for developing effective storage and monitoring systems, especially in agricultural contexts like rural Bangladesh. This analysis explores the biochemical pathways, microbial agents involved, and implications for crop quality assessment, providing a foundation for technological interventions like the "Smart Performance Monitoring System."

Ethanol

Microbes such as *Saccharomyces* (yeasts) and *Schizosaccharomyces* interact with the nutrients of fruits, particularly during ripening, converting glucose ($C_6H_{12}O_6$) and fructose into ethanol (C_2H_5OH) and carbon dioxide (CO_2) through fermentation.

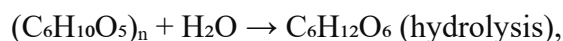


The reaction is a hallmark of anaerobic respiration in ripening fruits, where oxygen availability is limited (Jay, 2012). This process is especially prevalent in fruits like bananas and apples, where ethanol accumulation can indicate over-ripening or early spoilage, serving as a key indicator for quality monitoring systems.

Methane

The decomposition of complex carbohydrates such as starch and cellulose, represented as $(C_6H_{10}O_5)_n$, involves anaerobic bacteria like *Clostridium* and methanogenic archaea. These

microbes break down starch and cellulose into methane (CH₄) and CO₂ through a two-step process:



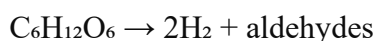
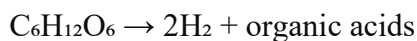
followed by



This pathway is significant in stored grains and root crops, where methane production signals advanced decomposition, posing challenges for long-term storage in humid environments.

Hydrogen

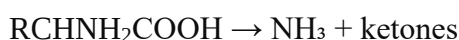
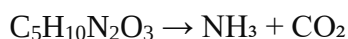
Hydrogen (H₂) production arises from anaerobic microbial processes, notably involving molds like *Rhizopus* and other microbes acting on glucose (C₆H₁₂O₆) or oleic acids.



Additionally, oleic acids, a common lipid in oilseeds, are metabolized by microbes to produce H₂ alongside aldehydes. Hydrogen accumulation is an early indicator of anaerobic conditions, which can accelerate spoilage in poorly ventilated storage facilities.

Ammonia

Protein decomposition, facilitated by bacteria, results in the release of ammonia (NH₃) and CO₂. The process involves the breakdown of amino acids (e.g., R-CHNH₂COOH) into NH₃, CO₂, and ketones through deamination. For instance, bacteria metabolize proteins to yield



depending on the substrate (Jay, 2012). This is particularly relevant for protein-rich crops like legumes, where ammonia detection can signal protein degradation and potential toxin formation, such as biogenic amines, which affect food safety.

Implications for Crop Storage and Quality Assessment

The production of these gases reflects the microbial degradation of crop nutrients, with each gas serving as a biomarker for specific stages of decomposition. Ethanol and CO₂ are early indicators of fruit ripening or fermentation, while methane and hydrogen suggest advanced anaerobic breakdown, often associated with mold or bacterial proliferation (Sanaeifar et al., 2017). Ammonia, linked to protein decay, indicates severe spoilage and potential health

hazards due to toxin production. These gaseous byproducts align with the environmental conditions outlined in prior discussions—high moisture (70–90%), warm temperatures (20–40°C), and poor ventilation—creating ideal conditions for microbial growth in rural storage settings (Kader, 2002).

Proper storage practices, such as maintaining low humidity and temperature control, can suppress these microbial activities, extending shelf life by months or years (Thompson, 2010). However, in resource-constrained regions like Bangladesh, such controls are often impractical, necessitating real-time monitoring systems. The detection of these gases using sensors (e.g., MQ-3 for ethanol, MQ-4 for methane, MQ-135 for ammonia) integrated with a microcontroller like the ESP32 offers a cost-effective solution.

3.1.3 Formulation of Problem

- The problem we are trying to address in this project is to assess the quality of crop yield (grains, fruits, vegetables) in a non-destructive and low-cost manner.
- To do so, we will be using:

Weighting Platform	BC547 BJT
Load Cell	DHT Humidity and Temp. Sensor
HX711 Amplifier	MQ-3 Gas Sensor (Ethanol)
ESP32	MQ-4 Gas Sensor (Methane)
12V Fan	MQ-8 Gas Sensor (Hydrogen)
RFID card reader	MQ-135 Gas Sensor (Ammonia)
OLED Screen	Push Button
Container for the yield	12 V Battery

3.1.4 Sensor analysis: Gas Sensors

An MQ sensor is a metal oxide sensor, also known as a chemiresistor, used to detect specific gases in the air. The core component is a sensing element, typically made of tin dioxide (SnO_2) coated on an aluminum oxide (Al_2O_3) ceramic substrate. SnO_2 has a high electrical resistance in clean air due to oxygen molecules adsorbing on its surface. The sensor includes a heater (usually a nickel-chromium coil) that heats the sensing element to an optimal temperature (200–300°C) for gas detection. When target gases interact with the SnO_2 surface, they react with adsorbed oxygen, releasing electrons back into the material. This reduces the sensor's resistance and the change is proportional to the gas concentration.

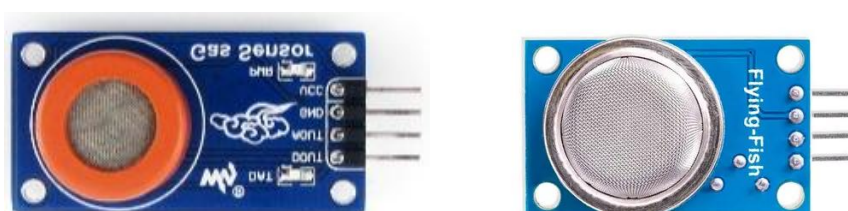


Figure: MQ-3 and MQ-4 gas sensor used for alcohol and methane

The sensor uses a voltage divider circuit with a load resistor to convert the resistance change into a measurable voltage signal. It provides:

- **Analog Output (AO):** A voltage (0–5V) proportional to gas concentration
- **Digital Output (DO):** A high (5V) or low (0V) signal based on a threshold set by a potentiometer, useful for simple detection without a microcontroller.

Different MQ models use specific doping materials to enhance sensitivity to target gases, but they may also respond to other gases, reducing specificity. They are not highly accurate without calibration and are best for detecting trends or thresholds rather than precise measurements.

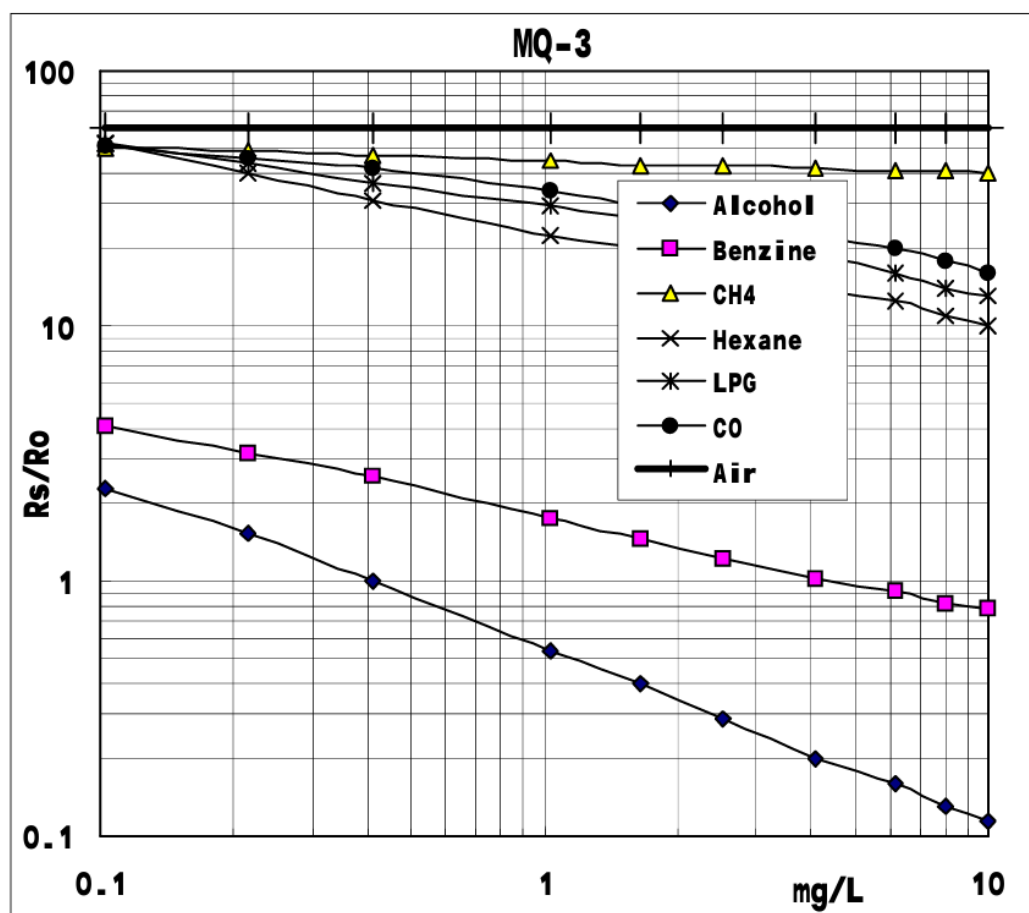


Figure: MQ-3 sensor sensitivity curve for different gasses which is used for converting the sensor data into ppm.

Gas sensor calibration:

The gas sensor from analog data, when biased between 3.3V and ground, gives a voltage output that ranges between 0V to 3.3V. This voltage data is reflective of the gas concentration near the surface of SnO₂, however, to find the exact concentration, we need to find a function that relates the voltage level into concentration. In our case, we shall convert the voltage data into ppm (parts per million). The calculation for voltage level to ppm conversion is shown below

for the MQ-3 sensor. The other sensor calibrations are done in the same way as well.

The sensitivity curve shows the data for mg/L concentration. Since 1 mol air = 24.45 L air,

$$ppm = \frac{mg/L \times 24450}{g/mol}$$

Now, the MQ-3 gas sensor is used to detect alcohol in this case. From many gas curves, the alcohol curve is the bottom most curve and shows the most sensitivity to concentration changes. 1 mol alcohol (ethanol in this case, C_2H_5OH) has a mass of 46 gm.

Two distinct points on the sensitivity curve are (0.1, 2.2) and (10, 0.12). Now, notice the curve is fairly linear in a log-log plot. Thus, $\log\left(\frac{R_s}{R_o}\right)$ vs $\log(conc.)$ has a linear relationship and can be approximated by the curve,

$$y = mx + c$$

The distinct points on the sensitivity curve, becomes (-1, 0.342) and (1, -0.921) in a logarithmic function. Using these points,

$$\begin{aligned} 0.342 &= -m + c \\ -0.921 &= m + c \end{aligned}$$

Solving, these two equations, we get, $c = -0.2895 \approx -0.29$ and $m = -0.6315 \approx -0.63$

So, the equation becomes,

$$\log\left(\frac{R_s}{R_o}\right) = -0.63\log(conc. mg/L) - 0.29$$

As the change in resistance causes an equivalent change in voltage, the equation becomes,

$$\log(V_{ratio}) = -0.63\log\left(\frac{ppm \times 46}{24450}\right) - 0.29$$

Finally,

$$ppm = 27.71 V_o^{1.587}$$

This format of,

$$\text{concentration} = a \times V_o^b$$

is also same for the other three sensors and are found from the sensitivity curve from the datasheet in a similar manner. The constant values for a, b for different gas sensors are shown in the table below:

Gas Sensor	Constant 'a' value	Constant 'b' value
MQ-3	27.71	1.59
MQ-4	35.89	2.83
MQ-8	5	2.5
MQ-135	0.3	2.85

Temperature & Humidity Sensor

The DHT22 (also known as AM2302) is a factory calibrated digital sensor that measures temperature with a NTC resistor and relative humidity with a moisture sensitive polymer dielectric capacitor. The thermistor's resistance decreases as temperature increases, allowing temperature calculation based on resistance changes. The capacitance changes with the amount of water vapor absorbed by the polymer, which correlates to relative humidity. An internal 8-bit microcontroller processes the analog signals from both sensors, converts them to digital values, and communicates the data to an external device. Resolution 0.1% RH and accuracy has a tolerance value of 2-5% RH and for temperature the resolution is 0.1°C and accuracy tolerance of 0.5°C with a range of -40°C to 80°C.



Figure: DHT22 sensor for temperature and relative humidity measurement

Weight Sensor

A bar-shaped load cell is a steel bar with strain gauges bonded to its surface. The strain gauges are thin, electrically conductive elements that change resistance when stretched or compressed. When weight is applied to the bar, via a platform or direct force, it bends or deforms slightly which causes the strain gauges to stretch or compress, altering their electrical resistance. The strain gauges are wired into a Wheatstone bridge circuit, which detects small changes in resistance. When the bar deforms, the bridge becomes unbalanced, producing a voltage proportional to the applied weight. The voltage output (typically in mV) is amplified and processed by an electronic circuit (HX711 amplifier).



Figure: A bar shaped load cell with HX711 amplifier

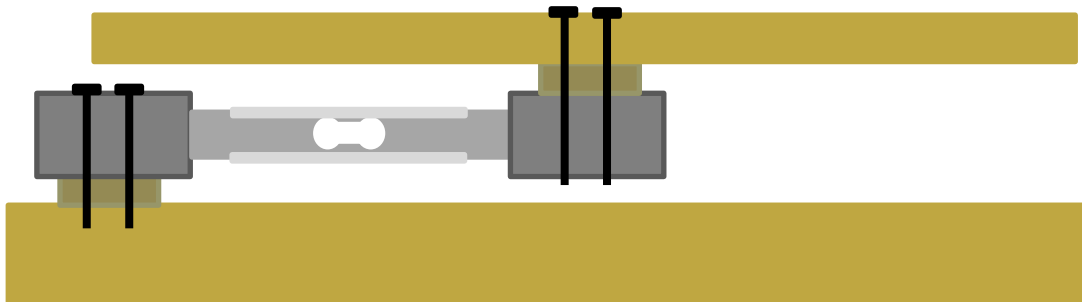


Figure: Design of weighing machine using load cell

To use the load cell for measurement of weight, we designed a weighing machine using a wooden structure. One end of the load cell is attached to the ground platform and the other end is connected to the upper platform where load will be placed causing bending in the bar.

RFID Sensor

An RFID (radio frequency identification) module identifies RFID tags using radio waves (usually at 13.56 MHz). It consists of an antenna and a transceiver. A microcontroller like ESP32 powers the antenna and generates an EMF around it (few cm to meters) allowing communication through SPI protocol. RFID tags are passive devices which are powered up by RFID module's electromagnetic field. The tags modulate the field to transmit its stored data (a unique id) and the RFID module decodes this data and sends it to ESP32 through SPI protocol.



Figure: MFRC522 RFID module

Output modules: OLED and Fan

The ESP32 sends commands and data to the OLED's driver IC (SSD1306), which controls the organic light-emitting diodes. It initializes the display, sets configurations (e.g., resolution, contrast), and sends pixel data to the display's memory, which the driver uses to illuminate specific pixels. A 12V brushless DC fan is used to decontaminate the storage air for newer readings which is driven by a 12V battery and is controlled using a switch implemented by a BJT (BC547).



Figure: OLED display and a 12V DC fan

Microcontroller ESP32

The ESP32 is a powerful, low-cost microcontroller with integrated Wi-Fi (2.4GHz 802.11) and Bluetooth (v4.2 BR/EDR and BLE), developed by Espressif Systems. It uses Dual-core Xtensa LX6 (up to 240 MHz) microprocessor. It has 520 kB of SRAM and 448 kB of ROM and external flash support up to 4 MB. It has 34 programmable GPIOs and supports ADC, DAC, SPI, I2C, I2S, UART etc. This microcontroller is used to connect everything together and send data to google sheet with the WiFi module.

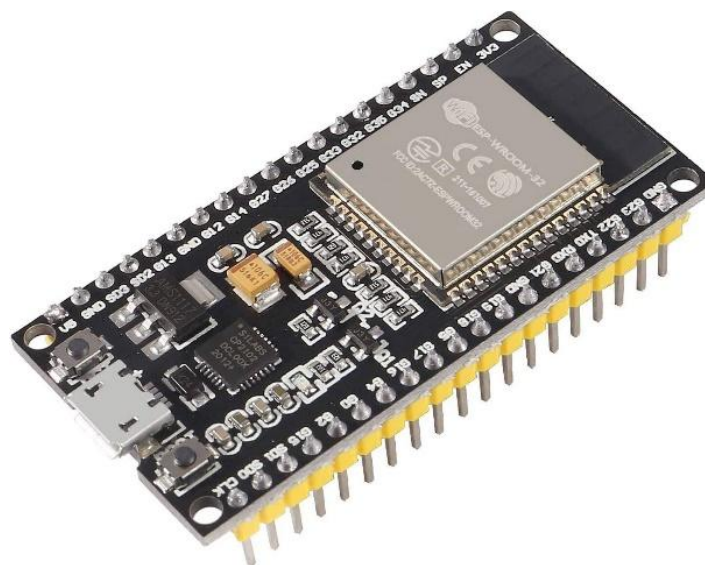


Figure: ESP32 microcontroller

3.2 Design Method (PO(a))

Block diagram

The following picture describes the whole design of our project's assembly.

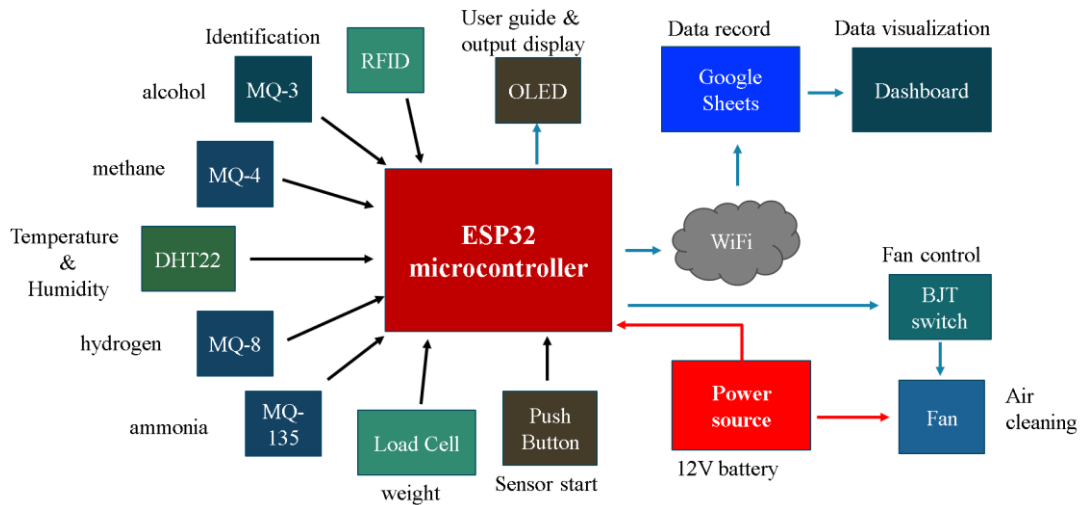


Figure: Layout of the whole Design

3.3 Pin Connection

The GPIO Pin connections are shown here.

ESP32 PIN	MODULE PIN	ESP32 PIN	MODULE PIN
GPIO 32	MQ-3 AO	GPIO 14	OLED SDA
GPIO 33	MQ-4 AO	GPIO 26	OLED SCL
GPIO 27	DHT22 DO	3V3	V+ rail
GPIO 36	MQ-8 AO	GND	V- rail
GPIO 35	MQ-135 AO	GPIO 18	RFID SCK
GPIO 4	HX711 DT	GPIO 23	RFID MOSI
GPIO 5	HX711 SCK	GPIO 19	RFID MISO
GPIO 22	RFID RST	GPIO 21	RFID SDA
GPIO 12	Fan (BJT Base)	GPIO 25	Push Button

HX711 PIN	Load Cell wire	ESP32 PIN	MODULE PIN
E+	Red Wire	A+	Green Wire
E-	Black Wire	A-	White Wire

3.4 Quality Assessment

Quality measurement of the crop yield is based on gas sensor readings in ppm/kg and environmental conditions like temperature and humidity.

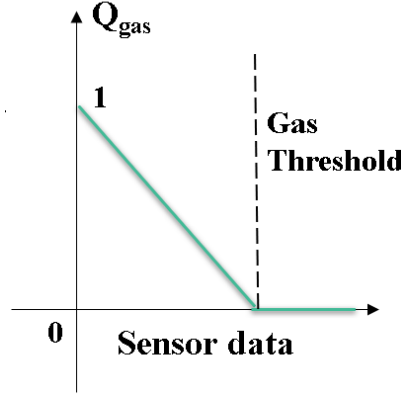


Figure: Quality Assessment of the yield w.r.t gas sensor data

Rotten fruit concentrations are taken as thresholds and quality index for fruit for each gas is defined as:

$$Q_{gas} = \max \left(1 - \frac{\text{Sensor reading}}{\text{Gas threshold}}, 0 \right)$$

Ideal storage humidity is 85-90% and ideal temperature is 0-15°C. Based on this information, we developed a quality index also based on temperature and humidity.

$$Q_{env} = 1 \text{ when in range}$$

$$Q_{env} = \max \left(1 - \frac{\text{Minimum} - \text{Sensor}}{\text{Range}}, 0 \right) \text{ when } \text{Sensor} < \text{Minimum}$$

$$Q_{env} = \max \left(1 - \frac{\text{Sensor} - \text{Maximum}}{\text{Range}}, 0 \right) \text{ when } \text{Sensor} > \text{Maximum}$$

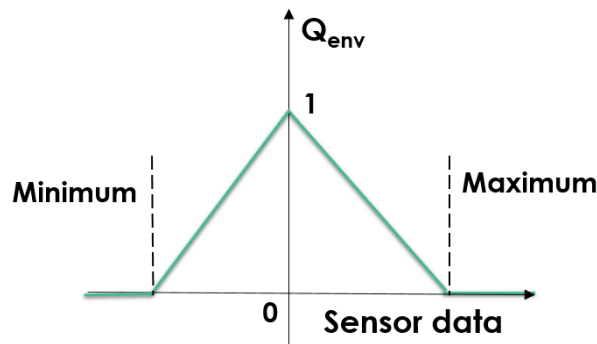


Figure: Quality Assessment of the yield w.r.t DHT sensor data

The overall quality index of the fruit is a weighted average of all the other quality indices.

$$Q_{index} = W_1 Q_{CH_4} + W_2 Q_{ethanol} + W_3 Q_{NH_3} + W_4 Q_{H_2} + W_5 Q_{temp} + W_6 Q_{hum}$$

Here, W_1 to W_6 are constants which should be derived with rigorous experimentations. With our simple experimentations with banana and mango, we defined the constants as,

$$W_1 = W_2 = \frac{2}{12}, W_3 = \frac{4}{12}, W_4 = \frac{2}{12}, W_5 = W_6 = \frac{1}{12}$$

4 Implementation

4.1 Description

Weighting Platform: Wooden platform capable of withstanding weight of upto 30-40 kgs. Designed to make sure weight can be accurately measured.



Figure: Weighting Platform

Load Cell: It uses Strain Gauge to measure weight of up to 50 kgs. We had to use extra screws to attach the load cell to the weighting platform.



Figure: Load Cell

HX711 Amplifier: The HX711 is a precision 24-bit ADC used to convert analog signals from a load cell into digital values for weight measurement.

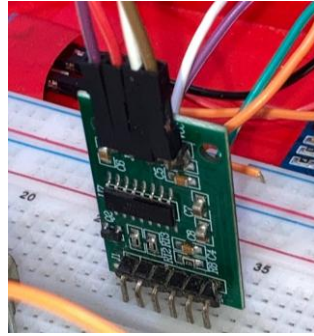


Figure: HX711 Amplifier

ESP32: Microcontroller with built-in WiFi capabilities.

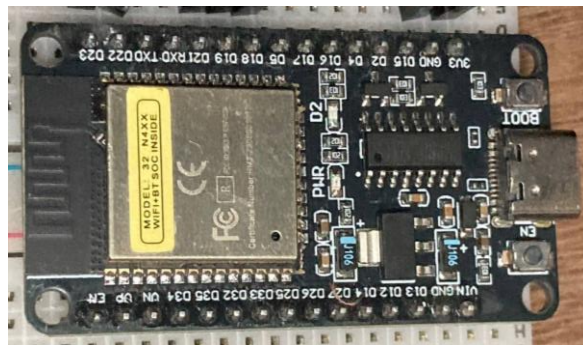


Figure: ESP-32 Microcontroller

RFID Card Reader: Reads ID stored on RFID cards using radio frequency signals.

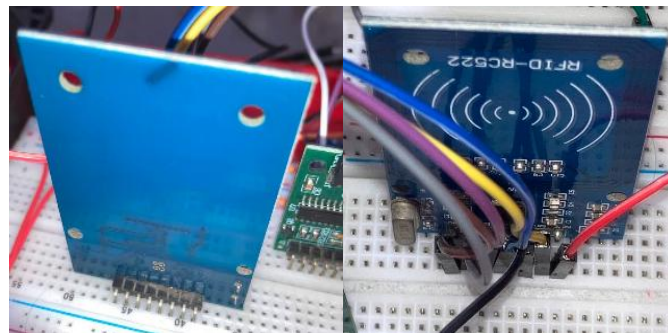


Figure: RDID – RC522 Card reader

Container: To contain the yield.



Figure: Yield Container

OLED: OLED to show different sensor output and interact with the user.

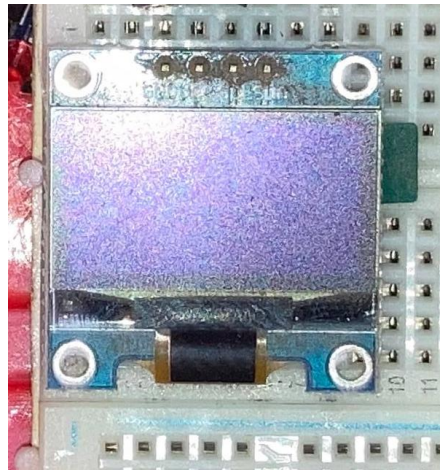


Figure: OLED

BC547 BJT: Works as a switch.

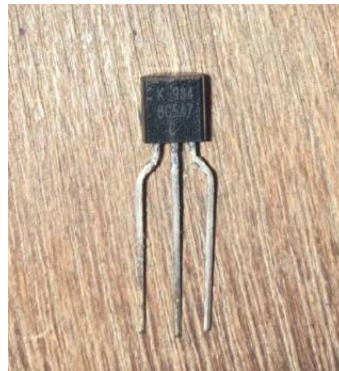


Figure: BC547 BJT

Fan: Exhaust fan activated to release gases once the yield is taken out of the container.



Figure: Exhaust Fan

Push Button: To take input from the user.



Figure: Push Button

Quality Monitoring Sensors (MQ & DHT): MQ-3, MQ-4, MQ-8 & MQ-135 to detect respectively Ethanol, Methane, Hydrogen & Ammonia coming out of the yield. DHT sensor is used to measure the temperature and the Humidity.

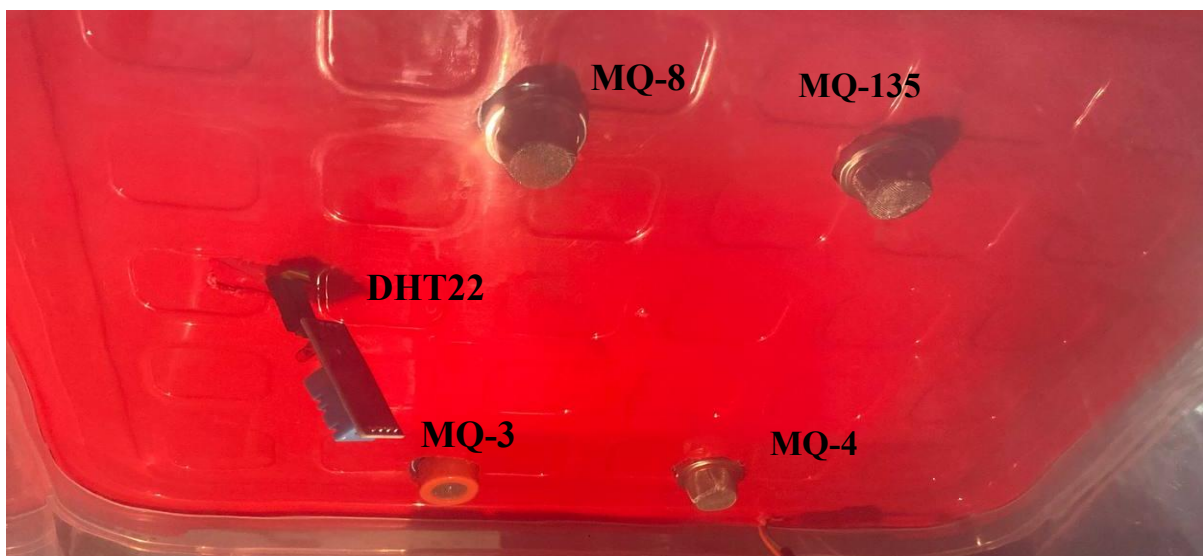


Figure: MQ and DHT sensors inside the container

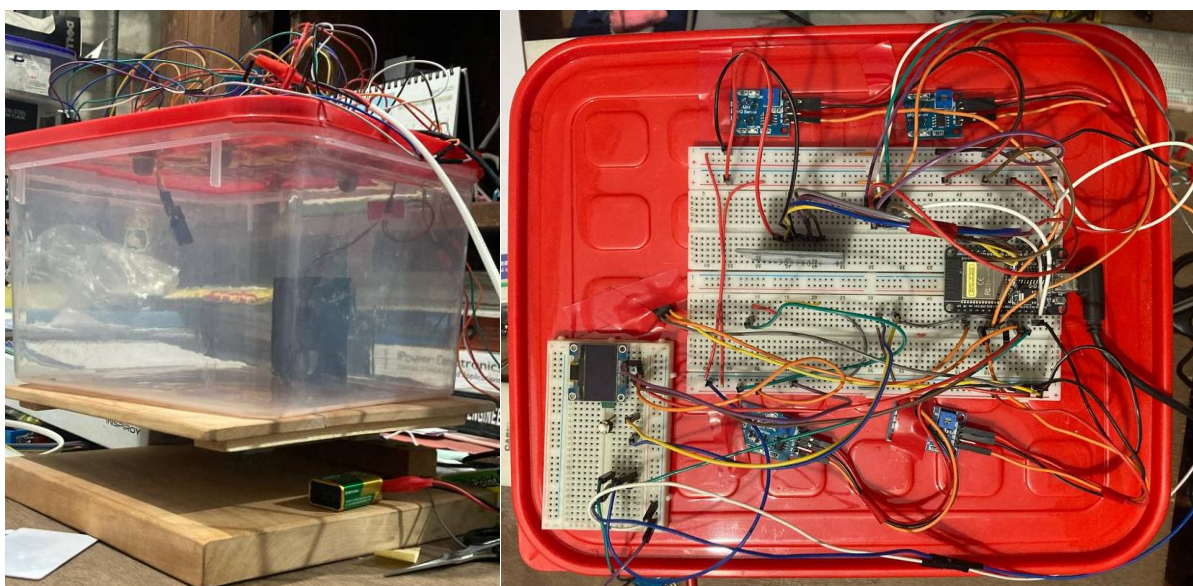


Figure: Complete Assembly

4.2 Process- Flow

Flowchart:

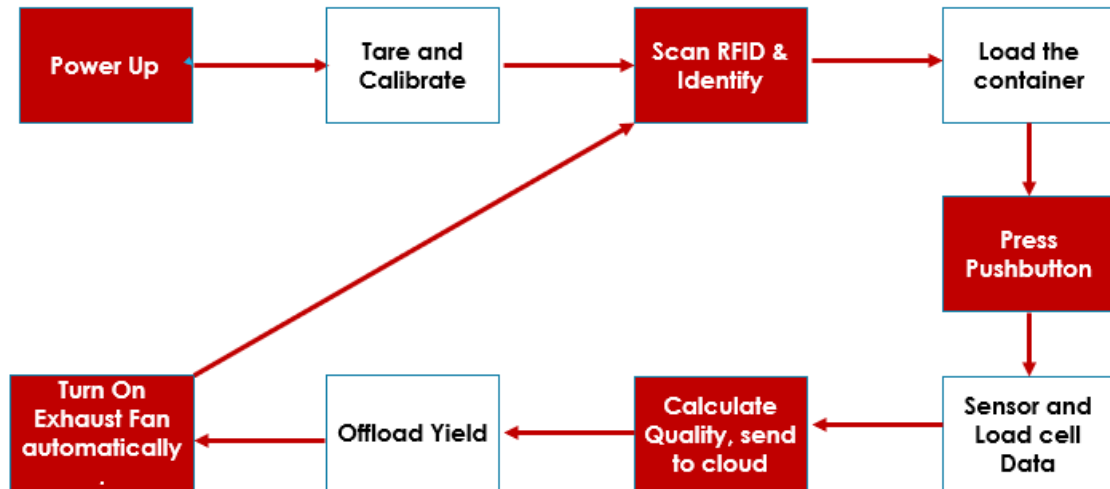


Figure: Flowchart

The system starts by powering up and calibrating sensors. Next, RFID identification is done, followed by loading the container and pressing the push button to trigger sensor data collection, calculation of quality indices, cloud upload, and finally automatic exhaust fan activation. Then the system again comes back to the Scan RFID state.

User's View:



Figure: User's View on the OLED screen

The OLED screen in our project serves the purpose of interacting with the user. It shows instructions to the user at different stages. Once the data is taken, the user is able to see quality of the yield in this OLED screen.

Data Logging:

Micro_project															
File Edit View Insert Format Data Tools Extensions Help															
100% 123 Default 10 B I A															
E19															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
1	User Name	Timestamp	Load Cell	NH3	CO2	CH4	Alcohol	Temperature	Humidity	Quality Index					
2	Anindya	24/07/2025 18:28:08	6.808233	1.181142	0.5195837	87.03401	30.52482	33	91	71%					
3	Anindya	24/07/2025 18:37:25	6.319867	1.255641	0.5367129	82.18914	28.54012	33	91	76%					
4	Ashik	24/07/2025 18:45:50	4.5148	1.382388	0.531467	89.08136	29.44999	33	90	69%					
5	Aritra	25/07/2025 22:55:45	2.100473	1.753256	0.5472908	122.5256	30.95876	31	92	44%					
6	Ashik	25/07/2025 22:59:18	2.104881	1.642623	0.538217	120.3652	31.48872	31	92	49%					
7	Aritra	25/07/2025 23:03:22	2.101691	1.663363	0.520322	121.4423	31.9591	31	91	59%					
8	Rohan	25/07/2025 23:13:24	2.103913	1.496897	0.4991464	113.1701	30.12393	31	91	73%					
9	Ashik	25/07/2025 23:14:42	2.106065	1.471508	0.5082583	120.7895	31.97908	31	91.5	31%					
10	Anindya	26/07/2025 08:23:38	1.675997	2.147922	0.4019402	130.8899	29.00668	31	94	56%					
11	Anindya	26/07/2025 08:29:16	2.226908	2.292619	0.405723	139.3831	31.6606	31	93	67%					
12	Nittya	26/07/2025 08:31:48	1.080112	2.251706	0.3984352	125.196	28.81953	31	93	66%					
13	Anindya	26/07/2025 08:34:22	8.351879	2.256413	0.3937006	125.1937	27.69296	31	92	59%					
14	Anindya	26/07/2025 14:01:18	2.093189	1.304254	0.3446961	103.2339	26.75875	31	90	48%					
15	Nittya	26/07/2025 14:02:33	2.095711	1.260591	0.3407287	103.0076	26.63833	32	88	82%					
16	Anindya	26/07/2025 14:10:34	4.8	0	0.3	34.8	21.3	30	85	86%					
17															

Figure: Continuous Data Logging into the Google Sheet

The built-in WiFi module of the ESP32 microcontroller sends the data from the sensors to the google sheet. So, the authority can always monitor the system. Also, this sheet works as an interconnect between the final dashboard and our ESP32 microcontroller.

Dashboard:

A dashboard for better visualization of data was created using html and JavaScript and deployed via GitHub pages. The html code ensured visualization and the JavaScript parsed the google sheet as a csv file and used the column values to process the data.

Features of the dashboard includes:

1. Summary of total sensor readings, total number of weights measured, best farmer of the week, best farmer all time based on highest quality, average temperature of storage
2. Latest reading data
3. Detailed performance analysis of individual farmer
4. Pie chart based on production weight
5. Quality trend as well as concentration of different gases with respect to time

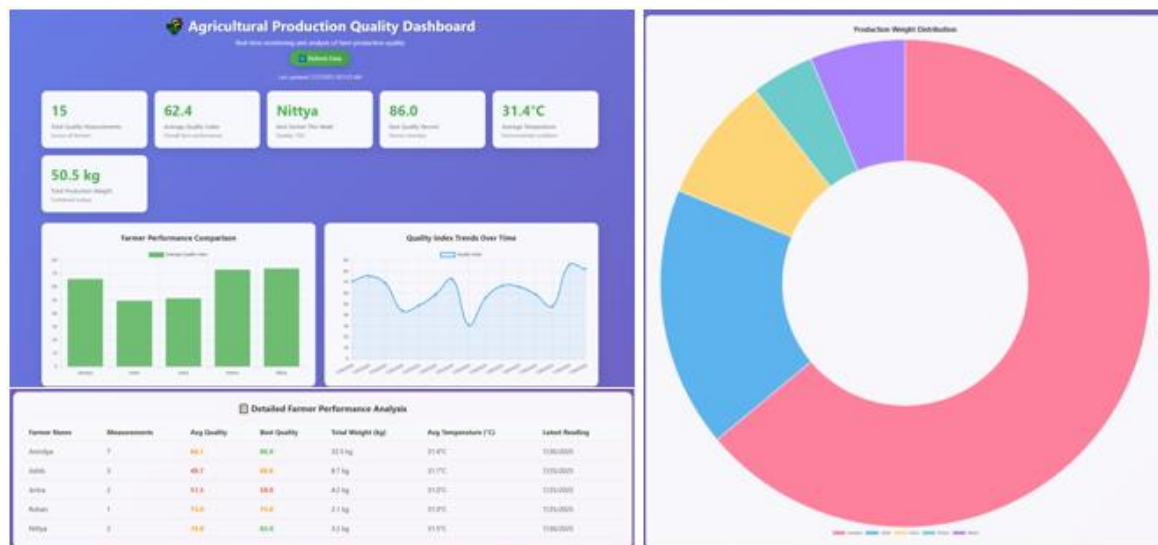


Figure: Data Presentation on the Dashboard

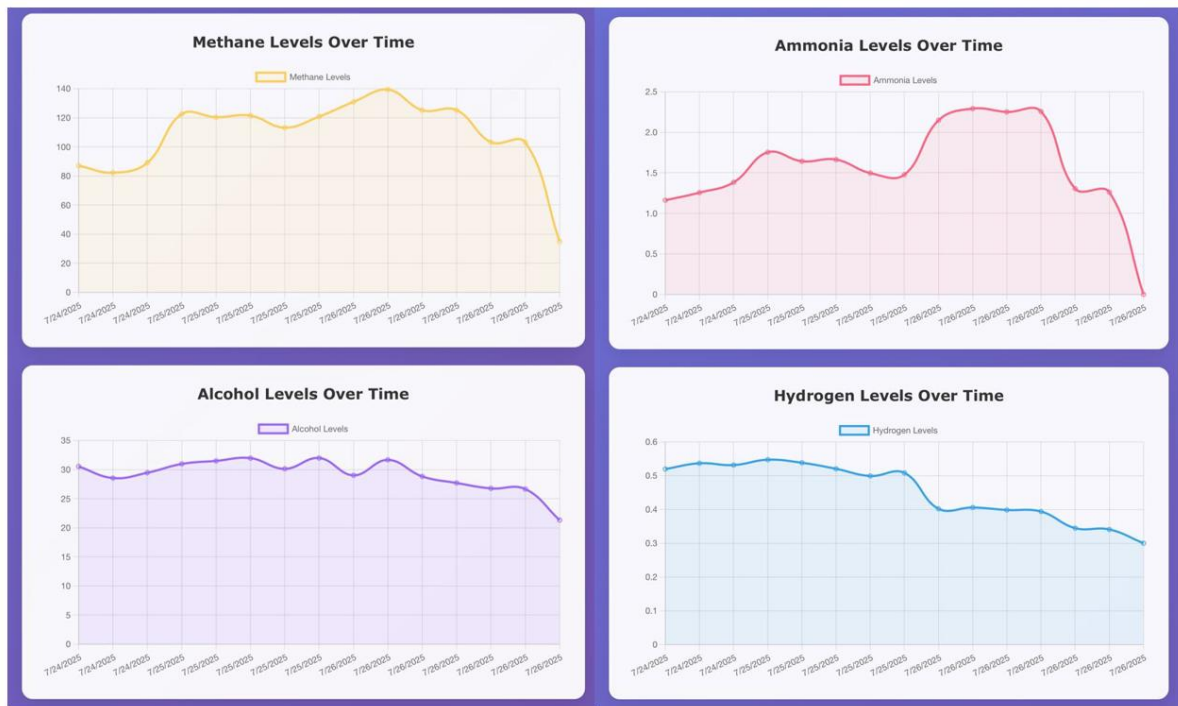


Figure: Continuous checking of the sensor data

4.3 Result & Analysis

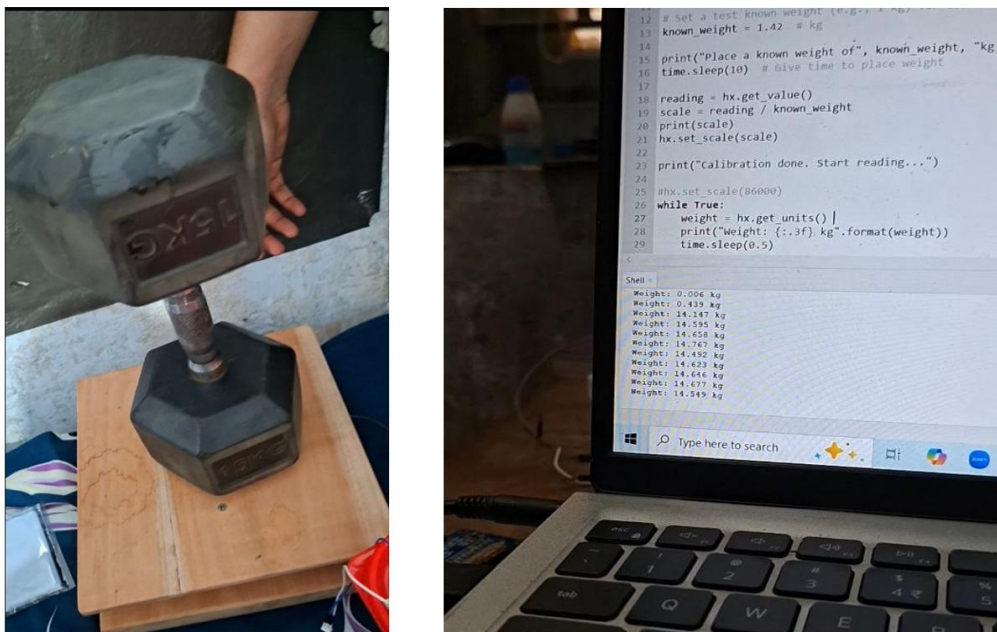


Figure: 15 Kg weight correctly measured by our Load Cell

Our project had two most important aspect. The first one was to weight at least 15 kgs and to process the sensor data to give verdict on the yield quality. For the first task, here we show how we were able to successfully measure 15 kg weight our Load Cell. Although we believe the load cell can weight up to 20-30 kg, we couldn't test it due to the unavailability of such huge weight in something small in size.

For our second part, we show sensor data for decomposed banana and overripe mango. In the following picture we can see concentration of different gases inside the container when we put these fruits. As we can see, for decomposed banana concentration of all gases are very high. Also the quality is very low. For overripe mango the sensor values are still high, but it is less than the concentrations for banana.

<pre> === Sensor Readings === Alcohol (MQ-3): 34.6 ppm Methane (MQ-4): 112.5 ppm Ammonia (MQ-135): 2.6 ppm Hydrogen (MQ-8): 46.1 ppm Temperature: 32 °C Humidity: 86 % Quality: 8.333333% ===== </pre>	<pre> === Sensor Readings === Alcohol (MQ-3): 28.4 ppm Methane (MQ-4): 107.1 ppm Ammonia (MQ-135): 1.1 ppm Hydrogen (MQ-8): 45.2 ppm Temperature: 32 °C Humidity: 88 % ===== </pre>
--	--

Figure: Sensor values for decomposed banana (L) & overripe mango (R)

Another noteworthy aspect from these snapshot is, the ammonia concentration is significantly higher for the decomposed banana that it is for overripe mango. The reason is ammonia is only released when the fruit is completely spoiled. As we discussed earlier, in our empirical weighted average formula, we gave Ammonia the highest weight. For further data analysis, a quick look into the dashboard for the project will help.

5 Analysis and Evaluation

5.1 Novelty

Combined RFID, load cell, gas, and humidity sensors in a single ESP32-based system. Enabled real time data monitoring using Google Sheets at minimal cost. Provided on-field feedback through OLED display—no need for smartphones. Tailored for local farming needs in Bangladesh. Modular design allows easy future upgrades.

5.2 Design Considerations

Considerations to public health and safety: It can detect spoilage using gas sensors, ensuring only safe produce is processed. It prevents exposure to harmful gases through early detection alerts.

Considerations to environment: It reduces excessive use of water and fertilizers through humidity monitoring. It minimizes post-harvest waste by identifying spoilage in real time. Also the low-power ESP32 system supports energy-efficient operation.

Considerations to cultural and societal needs: The Dashboard and UI are designed with rural farmers in mind, potentially in Bangla. It can increase transparency and accountability in agricultural labor practices. It can make IoT technology accessible to underdeveloped farming communities

5.3 Limitations of Tools (PO(e))

The gas and humidity sensors exhibited minor deviations over time, necessitating regular recalibration to maintain accuracy. The ESP32's limited onboard memory restricts the amount of local data logging without cloud support. Battery-powered operation imposes power constraints, requiring frequent charging or careful power management for extended deployments. Additionally, environmental factors such as wind, dust, and high humidity can cause fluctuations in sensor readings, reducing consistency. The system's reliance on Wi-Fi connectivity also poses challenges in rural areas with poor internet coverage, leading to delays in real-time data syncing.

5.4 Impact Assessment (PO(f))

Assessment of Societal and Cultural Issues: It promotes smart farming in rural areas, aiding small-scale farmers. It encourages adoption of sustainable technology in agriculture.

Assessment of Health and Safety Issues: It reduces health risks by enabling early detection of spoiled produce and promotes safer working conditions through effective monitoring systems.

Assessment of Legal Issues: It ensures the ethical use of collected data with minimal personal intrusion and aligns with guidelines for responsible technology use in rural communities.

5.5 Sustainability Evaluation (PO(g))

Assessment of Societal and Cultural Issues: The project promotes smart farming in rural areas, supporting small-scale farmers. It encourages the adoption of sustainable technologies to improve agricultural practices.

Assessment of Health and Safety Issues: It reduces health risks by enabling early identification of spoiled produce. Additionally, it fosters safer working environments through effective monitoring systems.

Assessment of Legal Issues: The system ensures ethical use of collected data with minimal personal intrusion. It aligns with established guidelines for responsible technology deployment in rural communities.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Name	ID	Contribution
Nittya Ananda Biswas	2006025	Design, Coding, Assembly
Ashik Shahriar	2006026	Individual Sensor Testing, Coding
Abul Fath	2006027	Debugging, Logistics
Kawsain Bin Salim	2006028	Logistics, Presentation, Report
Anindya Bhattacharjee	2006029	Design, Coding, Assembly, Dashboard

6.2 Log Book of Project Implementation

Date Range	Activities
14 Jun – 24 Jun	Planned system design, component selection, and team roles.
24 Jun – 04 Jul	Purchased sensors and components; built initial structure.
05 Jul – 08 Jul.	Tested and verified all sensors for functionality.
09 Jul – 16 Jul	Mounted sensors and completed coding for integration.
17 Jul – 27 Jul.	Debugged sensor data and ensured system stability.

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

In the project we developed a Smart Performance Monitoring System for agricultural farms using an ESP32 microcontroller integrated with RFID, load cell, gas, temperature, and humidity sensors. It monitor crop quality in real time, detecting spoilage and environmental conditions to reduce waste and improve storage practice. Data is sent via Wi-Fi to a cloud platform and displayed on an OLED screen for immediate farmer feedback. This low-cost, non-destructive system support sustainable farming in rural Bangladesh by enhancing productivity and safety. Despite some limitations, it offers an accessible solution for small-

scale farmers to make informed decisions and reduce post-harvest losses.

7.2 Github Link

The main code on which the esp32 runs is uploaded to GitHub as main.py and also the dashboard code is uploaded as index.html in the link shown below:

[Anindya-02/Agri_farm_production_quality: A dashboard for the data taken from the microprocessor and embedded systems project which assesses the quality of crop yields using different gas sensors](#)

7.3 Drive Link:

The drive link contains the demonstration videos of different sections of the system as well as the entire project.

https://drive.google.com/drive/folders/1u5YC6j6CkrjQr1M_FuJGyy4J8DGw6XMQ?usp=sharing

7.4 Dashboard Link:

Here is the dashboard link where we can see online visualization of readings of the sensors.

[Agricultural Production Quality Dashboard](#)

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Components	Price (Taka)
Weighting Platform	2000
Load Cell	600
HX711 Amplifier	80
ESP 32	450
12 V Fan	100
RFID card reader	120
OLED Screen	300
Container for the yield	250
BC547 BJT	5
DHT22 Sensor	70
MQ-3 Gas Sensor	120
MQ-4 Gas Sensor	150
MQ-135 Gas Sensor	160
Push Button	5
12 V Battery	800
Misc.	290
Total:	5400

9 Future Work (PO(I))

Future research should focus on calibrating sensors for specific crop types and environmental conditions, addressing durability in dusty, humid settings, and integrating data with mobile platforms to improve farmer accessibility. Combining gas monitoring with controlled atmosphere storage (modulating O₂ and CO₂ levels) could offer a hybrid solution, though cost remains a barrier (Thompson, 2010).

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